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Appendix Overview

The appendix is organized as follows:
Section A: Details of generative pipeline for the LoCoMo dataset.
Section B: Statistics of LoCoMo dataset, license for data release and annotator details.
Section C: Experimental setup and implementation details.
Section D: Additional results from evaluation on the LoCoMo benchmark.

A Generative Pipeline for LoCoMo

A.1 Persona

We assign unique persona statement p to each agent $\mathcal{L}_i$. For this, we select a range of initial persona statements p_c from the MSC dataset (Xu et al., 2022), each encompassing 4 to 5 sentences. We employ `gpt-3.5-turbo` as $\mathcal{M}$ to expand these into full persona statement p, conditioning $\mathcal{M}$ on the chosen statements p_c. The prompt used for converting a short list of speaker attributes from the MSC dataset (Xu et al., 2022) into a complete persona summary is presented in Fig. 5. We also use a single example of speaker attribute $\rightarrow$ persona summary as an in-context demonstration along with the prompt. A small selection of personas showcasing the diversity of speakers in the LoCoMo dataset is demonstrated in Fig. 5.

A.2 Temporal Event Graph

As outlined in Sec. 3.2, we use an iterative process for generating event graphs consisting of causally connected events based on a given persona summary. The base prompt for describing the constitution of the event graph, the nature of events and causal connections between events is shown in

Prompt

```
Let's write speaker descriptions from a given set of life attributes. Add crucial details in the persona about
the person such as their name, age, marital status, gender, job etc. Add additional details like names of
family/friends or specific activities, likes and dislikes, experiences when appropriate.
```

Examples of generated personas

Ellen is a 55 year old woman who lives in a beautiful Victorian house with her husband. She is a history buff and enjoys collecting antiques and dolls. In her free time, she loves exploring her house's huge attic which holds her collection of old video games. She also loves watching Kpop and taking jazz classes. Her hobbies have allowed her to develop a great sense of rhythm and appreciate all kinds of music. She's also a huge fan of interior design, and loves going out to find additional pieces for her home.

Paul is a 27 year old male who works freelance and lives alone. He enjoys staying active and frequently goes for late night walks. To reward himself for exercising, he often orders pizza, his favorite food. With no one to share it with, he doesn't mind eating several slices at once. He is also a big fan of Dream Theater and loves to listen to their music when out on his walks.

Kylie is a 19 year old female college student in her first year of college. She loves staying active by running, which she does often. She also has a part time job to support her studies. She is single and shares her space with her beloved cat. Kylie has a handful of friends and lively lifestyle.

Figure 5: **Prompt for persona statement (p) generation and examples of personas in LoCoMo.** The prompt used to generate expanded persona statements (p) from initial personas (p_c) for the virtual agents in our conversation generation pipeline (top) and select examples of persona statements present in the LoCoMo dataset.

Fig. 6. First, the base prompt is used along with the prompt for event graph initialization to generate three independent events relevant to a given personality. Then, the base prompt is combined with the prompt for the iterative generation of events to continue generating events that are caused by one or more of the events that are already present in the graph. See an example of a persona and the corresponding temporal event graph in Fig. 7. In the example, Jack aspires to be a hotel manager. Consequently, he enrolls in a hotel management course in July, and after three months, he expresses his excitement about the course on social media. In a similar vein, his passion for gaming results in an invitation from a well-known gaming company.

A.2.1 Virtual Agent Architecture

As outlined in Section 3.3, the virtual agents in our generative pipelines are composed of two mechanisms, *Reflect & respond* (Park et al., 2023) and *Image sharing & response*.

Reflect & respond. This mechanism operates over a combination of short-term and long-term memory. The short-term memory is a summary of a session that is conditioned on the summary from a previous session. See the prompt given to LLMs in our pipeline for generating summaries, and an example of a generated summary, in Fig. 8. The long-term memory is a database of *observations* about each speaker, that are essentially assertive statements about the speaker's persona and life. See the prompt given to LLMs in our pipeline for generating observations, and an example of observations extracted from a conversation, in Fig. 9. In practice, the conversation is annotated with turn IDs for each turn, and the model is also instructed to indicate the turn IDs that directly contribute to each observation. This allows us to keep track of the evidence when using observations as the context for RAG-based models used in our experiments (see Section 5).

Image sharing & response. See prompts for implementing image-sharing and image-response behaviors in Figure 10.

A.3 Human Filtering

Human annotators are instructed to edit the LLM-generated conversations in the following scenarios:

- Remove an image if it is not relevant to the current dialog or the conversation.
- Add context about an image to the current speaker's dialog if it is not discussed by them but the subsequent speaker has reacted to the image.
- Replace an image if it does not match the caption that was used to query for images.

Base prompt for event graph generation

```
Let's write a graph representing events that occur in a person's life based on a short summary of their
personality. Nodes represent events and edges represent the influence of past sub-events on a current event.
- The graph is represented in the form of a json list.
- Each entry is a dictionary containing the following keys: "event", “date", "caused_by", "id".
- The "event" field contains a short description of the event.
- The “date" field contains a date.
- The "id" field contains a unique identifier for the event.
- The "caused_by" field represents edges and is a list of "id" of existing events that have caused this event.
Events in the "caused_by" field should occur on dates before the event they have caused. Generate as many
causal connections as possible.
- An example of a causal effect is when the event "started a vegetable garden" causes "harvested tomatoes".
- Events can be positive or negative life events.
```

Additional prompt for event graph initialization

```
For the following input personality, generate three independent events E1, E2 and E3 aligned with their
personality. Events can be positive or negative life events and should reflect evolution in the person's
relationships, state of mind, personality etc.
```

Additional prompt for iterative generation of causal events

```
For the following input personality, generate new events that are caused by one or more EXISTING events. Events
can be positive or negative life events and should reflect evolution in the person's relationships, state of
mind, personality etc. Do not repeat existing sub-events. Start and end your answer with a square bracket.
```

Figure 6: **Prompts for temporal event graph generation.** The prompt used to generate complete personas for the LLMs in our conversation generation pipeline (top) and examples of personas present in the LoCoMo dataset.

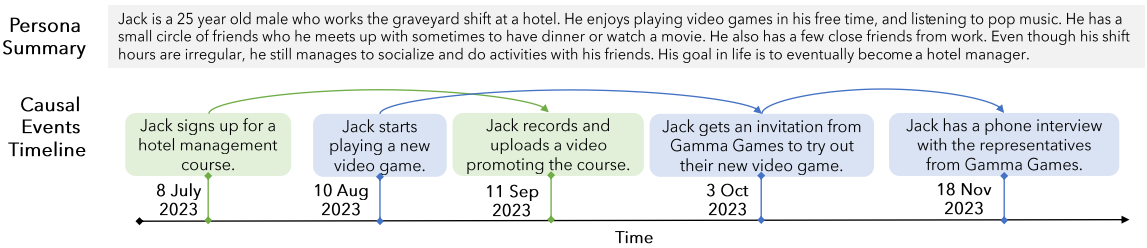

Figure 7: **Temporal Event Graph $\mathcal{G}$ Creation.** Each event is generated in accordance with the specified persona p and causal connections l between events are depicted to illustrate the casual relationships among them.

- Edit the dialog when the information present in the dialog is inconsistent with something said (or shared through an image) in earlier or later turns.
- Edit the dialog to ensure that the details in the conversation are consistent with those given in the event for the session.
- Remove any events from the event graph if they do not appear in the conversation.

See an example of some edits in Fig. 11.

B Dataset

B.1 Dataset Statistics

See a breakdown of the statistics of the conversations in the LoCoMo dataset in the top panel of Table 5. Also, see a breakdown of the statistics of the annotations in the evaluation benchmark in the bottom panel of Table 5.

B.2 Dataset License

The LoCoMo dataset will be released under the CC BY-NC 4.0 DEED license.[8]

B.3 Annotator Details

The annotators who worked on the LoCoMo dataset were in-house annotators and we were unable to obtain their demographics due to the confidential nature of such information.

C Experimental Setup

C.1 Baselines

The conversations in the LoCoMo dataset are composed of natural language dialogs and images that require higher-order reasoning and multimodal coreference resolution, respectively. From initial studies, we observed that multimodal coreference

[8] https://creativecommons.org/licenses/by-nc/4.0/